

LECTURE 2

Evolution of Computers — Before Electronic Computers

Learning Outcomes

- **Name and describe at least six calculating devices/machines used before electronic computers existed.**
- **Explain Charles Babbage's two major machines and why he is called the Father of Computers.**
- **Explain why Ada Lovelace is called the world's first computer programmer.**
- **Arrange all pre-electronic devices in correct historical/chronological order.**
- **Answer short, medium, and long-answer (up to 15 marks) examination questions on this topic with full explanations and diagrams.**

1. Why Study Old Calculating Devices?

Long before electronic computers existed, human civilisation still had a constant need to calculate — for trade and commerce, farming records, astronomy, navigation, taxation, and construction. Over many centuries, inventors created increasingly sophisticated tools and machines to make calculation faster, easier, and more reliable than doing it purely by hand or memory.

Studying this history is not just a matter of memorising names and dates for an exam — it helps us understand the gradual chain of ideas that eventually led to the modern electronic computer. Each device solved a specific limitation of the device before it, and this pattern of continuous improvement is a theme you will see repeated throughout the history of computing, including in the 'generations of computers' studied in the next lecture.

2. The Abacus (~3000 BCE)

The abacus is widely regarded as the oldest known calculating device in human history, believed to have originated in ancient Mesopotamia or China several thousand years ago. It consists of a wooden or metal frame containing a series of rods, with beads that can slide freely along each rod. Each rod typically represents a place value (units, tens, hundreds, and so on), similar to our modern decimal place-value system.

By sliding the beads according to a fixed set of rules, a skilled user can perform addition and subtraction, and with more advanced technique, even multiplication and division, remarkably quickly. The abacus does not perform any calculation automatically by itself — all the actual 'thinking' and rule-following is still done by the human operator; the abacus simply provides a fast, physical way to keep track of numbers.

Relevance today: The abacus is far from obsolete. Many small shopkeepers, particularly in parts of Asia, continue to use it for quick calculations even today. It also remains a popular teaching tool in primary schools around the world, including in India, to help young children build a strong intuitive understanding of place value (units, tens, hundreds) before they move on to purely mental or written arithmetic.

3. Napier's Bones (1617)

Napier's Bones was invented in 1617 by John Napier, a Scottish mathematician who is also famous for inventing logarithms. The device consisted of a set of specially numbered rods (traditionally made of bone or ivory, which is where the name comes from). Each rod represented the multiplication table of a single digit (0 through 9).

To multiply a large number, the user would select and arrange the appropriate rods side by side, and then simply read off partial results printed on the rods, adding them together diagonally to arrive at the final answer. This might sound complex today, but at the time, it was a major breakthrough, because multiplying large numbers by hand was slow, mentally exhausting, and highly prone to human error.

Napier's Bones considerably sped up trade calculations for merchants and land-measurement calculations for surveyors, and it remained in practical use in various forms for well over a century after its invention.

4. Pascaline (1642)

The Pascaline was invented in 1642 by the French mathematician and philosopher Blaise Pascal, who reportedly built it to help his father, a tax collector, with lengthy calculations. It is widely regarded as the first mechanical calculator — meaning it was the first device that could perform addition and subtraction automatically through mechanical means, rather than simply assisting a human who does the actual calculating.

The Pascaline worked using a system of interlocking gears and dials. Each dial represented a digit position (units, tens, hundreds, etc.), and the machine used a mechanism very similar to a modern vehicle's mechanical odometer: when one dial completed a full rotation (from 9 back to 0), it would automatically cause the next dial to advance by one position, exactly the way carrying works in addition.

Village Example: Think of the mechanical odometer (distance counter) in an old motorcycle or tractor. As the units-digit wheel completes a full turn, it automatically pushes the tens-digit wheel forward by one notch. The Pascaline used precisely this same gear-based 'carrying' idea, but applied it to performing addition of numbers rather than just counting distance.

5. Leibniz's Calculator (1673)

In 1673, the German mathematician and philosopher Gottfried Wilhelm Leibniz built upon Pascal's earlier design and created an improved mechanical calculator. While Pascal's machine could only perform addition and subtraction, Leibniz's machine introduced a clever mechanism known as the 'stepped drum' (or Leibniz wheel), which allowed the machine to also perform multiplication and division — the two operations Pascal's design could not handle directly.

This is a key exam point: Leibniz's Calculator is remembered specifically as the first mechanical calculator capable of performing all four basic arithmetic operations — addition, subtraction, multiplication, and division — within a single machine.

6. Charles Babbage — The Father of Computers

6.1 The Difference Engine (1822)

Charles Babbage, an English mathematician, philosopher, and inventor, designed the Difference Engine in 1822. His motivation came from a very practical problem of his time: mathematical tables (such as logarithm tables and navigational tables) were calculated and printed entirely by hand, by teams of human 'computers' (this is where the word 'computer' originally referred to a person who computes!). These manually prepared tables very often contained small but dangerous errors, which could cause serious mistakes in navigation, engineering, and science.

The Difference Engine was a large mechanical machine, built from thousands of precisely machined brass gears and levers, designed to automatically calculate and even print mathematical tables using a mathematical technique called the 'method of finite differences' — entirely removing the possibility of human calculation error. Although Babbage never completed a fully working, full-scale version of the Difference Engine in his own lifetime due to funding and engineering challenges, a complete working model was later built by London's Science Museum in the 1990s, based faithfully on his original drawings, proving that his design would indeed have worked.

6.2 The Analytical Engine (1837)

Babbage's true masterpiece of design, however, was the Analytical Engine, conceived in 1837. Unlike the Difference Engine, which could only calculate one specific type of mathematical table, the Analytical Engine was designed to be a general-purpose calculating machine — meaning it could, in principle, be given completely different instructions to solve many different kinds of mathematical problems, not just one fixed task.

The *Analytical Engine's design included two crucial parts that are conceptually identical to essential parts of a modern computer: a component called the 'mill', which performed the actual arithmetic calculations (functionally similar to today's CPU), and a component called the 'store', which held numbers waiting to be processed or already calculated (functionally similar to today's memory)*. It was also designed to accept its instructions through punched cards — a method of input borrowed from the automated textile looms of that era (the Jacquard loom), which used punched cards to control woven patterns in fabric.

Because of this revolutionary, general-purpose, programmable design — even though the machine itself was entirely mechanical and was never fully built during his lifetime — Charles Babbage is honoured today as the “Father of Computers”. This title is awarded not because he built a working electronic computer (he did not; electronic computers did not exist until roughly a century later), but because of the sheer originality and foresight of his DESIGN IDEAS, many of which are still fundamental to how computers are structured today.

7. Ada Lovelace — The First Programmer

Augusta Ada King, Countess of Lovelace, commonly known as Ada Lovelace, was an English mathematician who worked closely with Charles Babbage and deeply understood the potential of his Analytical Engine, perhaps even more clearly than Babbage himself did at the time.

While translating an Italian mathematician's article about the Analytical Engine into English, Ada Lovelace added her own extensive notes — which ended up being several times longer than the original article itself. Within these notes, she wrote out a detailed, step-by-step set of instructions describing exactly how the Analytical Engine could be used to calculate a sequence of numbers known as Bernoulli

numbers. This detailed set of instructions is widely considered to be the world's first computer program, or algorithm, ever written.

Why she is remembered: Even though the Analytical Engine itself was never actually built as a complete, working machine during her lifetime, Ada Lovelace's work proved something profoundly important: that a machine could be given a precise, logical, step-by-step sequence of instructions to solve a problem, and that such machines might one day be used for far more than simple number-crunching. This core insight — that machines can be 'programmed' with logical instructions — is the foundational idea behind all computer programming that exists today, which is why the modern programming language 'Ada' was named in her honour.

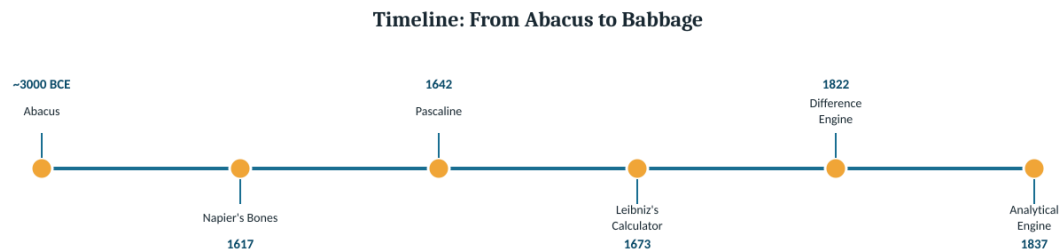


Fig 2.1 — Timeline of Pre-Electronic Calculating Devices

8. Common Doubt

Q. Was Babbage's Analytical Engine an electronic computer, like the ones we use today?

A. No, and this is a very important distinction to remember for exams. The Analytical Engine was entirely mechanical — built from gears, levers, rods, and wheels, with absolutely no electricity involved anywhere in its design. It is considered historically important because of its DESIGN IDEA (a general-purpose, programmable machine with separate processing and memory-like units), not because of the technology it used. True electronic computers, using vacuum tubes, would not appear until more than a hundred years later, starting in the 1940s — the topic of our next lecture.

9. Comparison Table of Pre-Electronic Devices

Device	Inventor	Year	What It Could Do
Abacus	Unknown (ancient)	~3000 BCE	Addition, subtraction (manual, bead-based)
Napier's Bones	John Napier	1617	Multiplication, division (using rods)
Pascaline	Blaise Pascal	1642	Addition, subtraction (mechanical, gears)
Leibniz's Calculator	G. W. Leibniz	1673	All 4 operations (+, -, ×, ÷)
Difference Engine	Charles Babbage	1822	Automatic calculation & printing of tables
Analytical Engine	Charles Babbage	1837	General-purpose, programmable via punched cards

10. Fill in the Blanks

- 1. The oldest known calculating device, using beads on rods, is the _____.
- 2. _____ invented a set of numbered rods in 1617 to assist with multiplication and division.
- 3. The Pascaline, invented by _____ in 1642, could perform addition and subtraction.
- 4. The first mechanical calculator to perform all four basic arithmetic operations was invented by _____.
- 5. Charles Babbage designed the _____ Engine in 1822 to automatically calculate mathematical tables.
- 6. The Analytical Engine used _____ cards as a method of input, borrowed from weaving looms.
- 7. The two key parts of the Analytical Engine were called the 'mill' and the '_____ '.
- 8. Charles Babbage is known as the _____ of Computers.
- 9. _____ is credited with writing the world's first computer program/algorithm.
- 10. The Analytical Engine was a _____ machine, not an electronic one.

Answers: 1. Abacus | 2. John Napier | 3. Blaise Pascal | 4. Gottfried Wilhelm Leibniz | 5. Difference | 6. punched | 7. store | 8. Father | 9. Ada Lovelace | 10. mechanical

11. True or False

- 1. The abacus is no longer used anywhere in the world today.
- 2. Napier's Bones could perform multiplication and division.
- 3. The Pascaline could perform all four basic arithmetic operations.
- 4. Charles Babbage's Analytical Engine was successfully built and used widely during his lifetime.
- 5. The Analytical Engine's 'mill' is conceptually similar to a modern CPU.
- 6. Ada Lovelace is remembered for building the first electronic computer.
- 7. The Analytical Engine used electricity to function.

Answers with Explanation

1. False. The abacus is still used by shopkeepers in parts of Asia and as a teaching tool in many primary schools.

2. True. It was specifically designed to assist with these two operations using numbered rods.

3. False. The Pascaline could only perform addition and subtraction; Leibniz's Calculator later added multiplication and division.

4. False. It was never fully built during his lifetime due to funding and engineering challenges; a working model of the Difference Engine was only built much later, in the 1990s.

5. True. The mill performed the actual calculations, similar to how the CPU processes data in a modern computer.

6. False. She is remembered for writing the world's first computer program (algorithm) for the Analytical Engine, which was a mechanical, not electronic, machine.

7. False. It was an entirely mechanical machine, made of gears, levers, and rods, with no electricity involved.

12. Short Answer Questions (2–3 Marks)

- Who invented the Pascaline? (2 marks)
- Who is known as the Father of Computers? (2 marks)
- In which year did Babbage design the Analytical Engine? (2 marks)
- What was Napier's Bones used for? (3 marks)
- Why is Ada Lovelace remembered in computer history? (3 marks)

13. Long Answer Model Answers (5, 10 & 15 Marks)

Q1 (5 Marks): Describe any four calculating devices used before electronic computers, in chronological order.

Before the invention of electronic computers, several important calculating devices were developed over many centuries to make calculation faster and more reliable.

The Abacus, dating back approximately 3000 years, is the oldest known calculating device, consisting of a frame with beads on rods used to perform addition and subtraction by a skilled human operator.

Napier's Bones, invented by John Napier in 1617, was a set of numbered rods that assisted users in performing multiplication and division by arranging the rods and reading off the results.

The Pascaline, invented by Blaise Pascal in 1642, was the first mechanical calculator, using gears to automatically perform addition and subtraction, with a carrying mechanism similar to a vehicle's odometer.

Leibniz's Calculator, built in 1673 by Gottfried Leibniz, improved upon Pascal's design by adding the ability to perform multiplication and division, making it the first machine capable of all four basic arithmetic operations.

Together, these devices show a clear, gradual progression from simple manual counting tools towards increasingly automatic mechanical calculation, setting the stage for Charles Babbage's later, far more ambitious designs.

Q2 (10 Marks): Explain the contribution of Charles Babbage to the field of computers, and describe the key features of his two major machines.

Charles Babbage, an English mathematician and inventor of the nineteenth century, is honoured today as the “Father of Computers” because of the extraordinary originality of his design ideas, even though he did not build an electronic computer — electronic computers did not exist until roughly a century after his death.

Babbage's motivation stemmed from a real and serious problem of his time: mathematical tables used for navigation, engineering, and science were calculated and printed by hand, and frequently contained dangerous errors due to human mistakes. He wanted to build a machine that could eliminate this human error entirely.

His first major design, the Difference Engine (1822), was a large mechanical machine built from thousands of precisely machined gears, intended to automatically calculate and print mathematical tables using a technique called the method of finite differences. Although a complete version was never finished in his lifetime due to funding and engineering limitations, a fully working model was later constructed by London's Science Museum in the 1990s, based on his original drawings, proving that his design was fundamentally sound.

His second and far more significant design was the Analytical Engine (1837), which was intended to be a general-purpose, programmable calculating machine, rather than one limited to a single fixed task. It featured two components that are conceptually similar to parts of a modern computer: the 'mill', which performed calculations (similar to today's CPU), and the 'store', which held data (similar to today's memory). It was designed to accept instructions through punched cards, an idea borrowed from the Jacquard weaving loom.

It is specifically this programmable design of the Analytical Engine — with its separation of processing and storage, and its ability to be given different instructions for different tasks — that earns Babbage his title as the Father of Computers. His work closely involved Ada Lovelace, who wrote what is considered the world's first computer program for this machine, further extending the significance of Babbage's design.

In conclusion, Charles Babbage's true contribution to computing lies not in any physical machine he successfully completed, but in conceiving, over a century before electronic computers existed, the essential architectural idea of a programmable machine with separate units for processing and storage — an idea that remains central to computer design even today.

Q3 (15 Marks): Trace the complete evolution of calculating devices from the Abacus to Babbage's Analytical Engine, describing each device, its inventor, its capability, and its historical significance, and explain the contribution of Ada Lovelace.

The history of calculating devices stretches back thousands of years, reflecting humanity's continuous effort to make calculation faster, more reliable, and less dependent on error-prone human memory. This answer traces this evolution in full chronological order.

The earliest known calculating device is the Abacus, dating back approximately 3000 years, consisting of a frame with beads that slide along rods representing place values. While it required a skilled human operator to perform the actual calculation, it provided a fast physical way of tracking numbers, and remains in limited use even today, including as a teaching tool for place value in primary schools.

In 1617, the Scottish mathematician John Napier invented Napier's Bones, a set of specially numbered rods that could be arranged to assist with multiplication and division, significantly speeding up trade and land-measurement calculations compared to purely manual methods.

In 1642, the French mathematician Blaise Pascal invented the Pascaline, widely regarded as the first true mechanical calculator, because it could perform addition and subtraction automatically through a system of interlocking gears, using a 'carrying' mechanism similar to a vehicle's mechanical odometer.

In 1673, the German mathematician Gottfried Wilhelm Leibniz improved upon Pascal's design with his own calculator, which added the ability to perform multiplication and division through a stepped-drum mechanism, making it the first machine capable of performing all four basic arithmetic operations.

The most significant advances, however, came from the English inventor Charles Babbage in the nineteenth century. In 1822, motivated by the frequent errors found in hand-calculated mathematical tables, he designed the Difference Engine, a mechanical machine intended to automatically calculate and print such tables without human error, using the method of finite differences.

- (A labelled timeline diagram, from Abacus through to the Analytical Engine, should be drawn here, exactly as shown in Fig 2.1 of these notes, to earn full diagram marks.)

Babbage's most visionary design, however, was the Analytical Engine of 1837 — a general-purpose, programmable machine, rather than one limited to a single task. It contained a 'mill' for performing calculations (similar in concept to a modern CPU) and a 'store' for holding data (similar in concept to modern memory), and it was designed to accept instructions via punched cards, an idea borrowed from

the Jacquard weaving loom. Although never fully built in his lifetime, this design is why Babbage is honoured as the Father of Computers.

Working closely with Babbage was Ada Lovelace, an English mathematician who deeply understood the potential of the Analytical Engine. While translating a description of the machine, she added extensive notes of her own, including a detailed, step-by-step set of instructions for calculating a mathematical sequence known as Bernoulli numbers. This is considered the world's first computer program, or algorithm, and it demonstrated the profound insight that machines could be given precise logical instructions to solve problems — the foundational idea behind all of modern programming.

In conclusion, the evolution from the Abacus to the Analytical Engine represents a continuous, centuries-long chain of innovation, with each device solving a specific limitation of the one before it — from simple manual bead-counting, through mechanical gear-based automatic calculation, and finally to Babbage's visionary, programmable, general-purpose design, brought to life conceptually by Ada Lovelace's pioneering work in programming. This entire chain of mechanical innovation set the essential intellectual foundation upon which the electronic computers of the twentieth century, studied in our next lecture, would eventually be built.

14. Exam Tips

- Remember the correct chronological order: Abacus → Napier's Bones → Pascaline → Leibniz's Calculator → Difference Engine → Analytical Engine.
- Always mention BOTH of Babbage's machines (Difference Engine AND Analytical Engine) in any answer about his contribution — many students mention only one and lose marks.
- For Ada Lovelace questions, the key phrase examiners look for is “first computer program / algorithm.”
- Draw the timeline diagram (Fig 2.1) for any 10 or 15 mark question on this topic — it demonstrates a clear understanding of chronology and earns dedicated diagram marks.
- Common Mistake: Do not call Babbage's Engines “electronic” computers — they were entirely mechanical. This is a frequently tested distinction.

References

- **Refer to your university-prescribed/standard Computer Fundamentals textbook for this topic, as per syllabus official reading list.**
- **Class lecture discussion and board work for this lecture.**